

2.4.1 Continuity Worksheet

1. Which one of the following must hold in order for a function g to be continuous at a number k ?

(a) $\lim_{x \rightarrow \infty} g(x) = g(k)$

(b) $\lim_{x \rightarrow \infty} g(x) = 0$

(c) $\lim_{x \rightarrow \infty} g(x) = k$

(d) $\lim_{x \rightarrow 0} g(x) = g(k)$

(e) $\lim_{x \rightarrow 0} g(x) = 0$

(f) $\lim_{x \rightarrow 0} g(x) = k$

(g) $\lim_{x \rightarrow k} g(x) = g(k)$

(h) $\lim_{x \rightarrow k} g(x) = 0$

(i) $\lim_{x \rightarrow k} g(x) = k$

2. State all values of x where the graph of g has a discontinuity. State the type of discontinuity for each.

$$g(x) = \frac{x^2 - 2x - 8}{2x^2 - 9x + 4}$$

We notice that $g(x) = \frac{x^2 - 2x - 8}{2x^2 - 9x + 4} = \frac{(x-4)(x+2)}{(2x-1)(x-4)}$

We can see that g is discontinuous at the following values of x :

$$x = \frac{1}{2} \text{ (infinite discontinuity)}$$

$$x = 4 \text{ (removable discontinuity)}$$

(because the graph has a hole at $x = 4$)

(because the graph has a VA of $x = \frac{1}{2}$)

3. Show that the function $f(x) = \begin{cases} \ln(x) + 1 & \text{if } x \leq e \\ (x-e)^2 + 2 & \text{if } x > e \end{cases}$ is continuous at $x = e$.

We see that:

$$\lim_{x \rightarrow e^-} f(x) = \lim_{x \rightarrow e^-} [\ln(x) + 1] = \ln(e) + 1 = 1 + 1 = 2$$

$$\lim_{x \rightarrow e^+} f(x) = \lim_{x \rightarrow e^+} [(x-e)^2 + 2] = (e-e)^2 + 2 = 0 + 2 = 2$$

Therefore $\lim_{x \rightarrow e} f(x) = 2$.

We also note that $f(e) = \ln(e) + 1 = 1 + 1 = 2$

So, $\lim_{x \rightarrow e} f(x) = f(e)$. Therefore $f(x)$ is continuous at $x = e$.

4. At which points is the following function discontinuous. Justify your answer.

$$f(x) = \begin{cases} (x-1)^2 & \text{if } x \geq 0 \\ x+3 & \text{if } -2 < x < 0 \\ (x+2)^2 + 1 & \text{if } x \leq -2 \end{cases}$$

Since each piece of the function f is a polynomial, we know that it is continuous on its given domain. The only points at which $f(x)$ may be discontinuous would be at $x = 0$ and $x = -2$ (these are the points where the function could break).

We see that $f(0) = (0-1)^2 = 1$.

We also note that $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x-1)^2 = (0-1)^2 = 1$ and

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x+3) = 0+3 = 3$$

Since $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$, we now that $\lim_{x \rightarrow 0} f(x)$ does not exist and therefore, f must be discontinuous at $x = 0$.

We see that $f(-2) = (-2+2)^2 + 1 = 1$.

We also note that $\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (x+2)^2 + 1 = (-2+2)^2 + 1 = 1$ and

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x+3) = -2+3 = 1$$

Since $\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^-} f(x) = 1$, we know that $\lim_{x \rightarrow -2} f(x) = 1$, which means that $\lim_{x \rightarrow -2} f(x) = f(-2)$.

Therefore, f must be continuous at $x = -2$.

So we see that f is only discontinuous at the point $x = 0$.

5. Show that $f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ \sqrt{x} & \text{if } x \geq 0 \end{cases}$ is continuous over $(-\infty, \infty)$.

f is continuous on $(-\infty, 0)$ because x^2 is a polynomial.

f is continuous on $(0, \infty)$ because $\sqrt{x}$ is a radical function defined on this interval.

We must check that f is continuous at $x = 0$.

We see that $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} x^2 = (0)^2 = 0$ and $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \sqrt{x} = \sqrt{0} = 0$

Therefore, $\lim_{x \rightarrow 0} f(x) = 0$. We also note that $f(0) = \sqrt{0} = 0$. So $\lim_{x \rightarrow 0} f(x) = f(0)$ and f is continuous at $x = 0$.

Therefore, f is continuous over $(-\infty, \infty)$.

6. Show that that $f(x) = \begin{cases} \sin(x) & \text{if } x < \pi/4 \\ \cos(x) & \text{if } x \geq \pi/4 \end{cases}$ is continuous over $(-\infty, \infty)$.

f is continuous on $(-\infty, \frac{\pi}{4})$ because $\sin(x)$ is a trig. function which is always defined on this interval.

f is continuous on $(\frac{\pi}{4}, \infty)$ because $\cos(x)$ is a trig. function which is always defined on this interval.

We must check that f is continuous at $x = \frac{\pi}{4}$.

We see that $\lim_{x \rightarrow \frac{\pi}{4}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^-} \sin(x) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ and $\lim_{x \rightarrow \frac{\pi}{4}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{4}^+} \cos(x) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

Therefore, $\lim_{x \rightarrow \frac{\pi}{4}} f(x) = \frac{\sqrt{2}}{2}$. We also note that $f\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$. So $\lim_{x \rightarrow \frac{\pi}{4}} f(x) = f\left(\frac{\pi}{4}\right)$ and f is

continuous at $x = \frac{\pi}{4}$.

Therefore, f is continuous over $(-\infty, \infty)$.

7. Find a value for a such that the function $f(x) = \begin{cases} x & \text{if } x \leq a \\ \sqrt{x} & \text{if } x > a \end{cases}$ is continuous at a .

We need to select a value of a such that $\lim_{x \rightarrow a} f(x) = f(a)$.

In order for this to be true we need $\lim_{x \rightarrow a} f(x)$ to exist. We note that

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^-} x = a \quad \& \quad \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} \sqrt{x} = \sqrt{a}$$

If $\lim_{x \rightarrow a} f(x)$ is going to exist we must select a value of a such that $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \rightarrow a = \sqrt{a}$.

Therefore, if we select $a = 0$ or $a = 1$ then $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} f(x) = a$.

We also know that $f(a) = a$ and therefore f is continuous at $x = a$ by definition.

8. Determine the values of c and d so that $f(x) = \begin{cases} 5 + x^2 & \text{if } x < -1 \\ cx + d & \text{if } -1 \leq x \leq 5 \\ 7x - 11 & \text{if } x > 5 \end{cases}$ is continuous over its domain.

Since each piece of the function f is a polynomial, we know that it is continuous on $(-\infty, -1) \cup (-1, 5) \cup (5, \infty)$. If $f(x)$ is to be continuous on its entire domain we need to look at the points of the graph where $x = -1$ and $x = 5$ (these are the points where the function could break).

We note that:

$f(-1) = c(-1) + d$ is defined and

$f(5) = c(5) + d$ is defined and

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} (cx + d) = c(-1) + d$$

$$\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} (7x - 11) = 7(5) - 11 = 24$$

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} (5 + x^2) = 5 + (-1)^2 = 6$$

$$\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} (cx + d) = c(5) + d$$

In order to be continuous at $x = -1$ we need

In order to be continuous we need

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^-} f(x) = f(-1), \text{ so we need}$$

$$\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^-} f(x) = f(5), \text{ so we need}$$

$$c(-1) + d = 6$$

$$c(5) + d = 24$$

Therefore, we have generated two equations:
$$\begin{cases} c(-1) + d = 6 \\ c(5) + d = 24 \end{cases}$$

By solving the system we find that we need $c = 3$ and $d = 9$.